

MARKING RUBRIC

Integrative Bioinformatics project assessment

Total weight: 100%

MARKING RUBRIC DOMAIN / SCORE	UNATTAINED score 0	SUFFICIENTLY ATTAINED score 6	EXCELLED score 1	WEIGHT 100%
Project Scheduling	- Did not deliver the initial Gantt chart - Failed to update the live Gantt chart by the end of the project	- Delivered the initial Gantt including most key identified tasks in scope - Succeeded in identifying most key task dependencies - Delivered the final Gantt reflecting real project status by the project deadline	- Delivered both Gantts successfully as expected in content and timing, with dependencies identified - Correct retrospective analysis of gaps between initial and final Gantt is present - Contingency plan for risk management of critical parts of the project are included in the Gantt (ideally first Gantt) - Live Gantt has been correctly maintained throughout project development	10%
Evidence gathering	- Information sources used are obsolete, inaccurate or untrustworthy - References used are irrelevant or only lightly related to the statements they support - Reference formatting applies a wrong format or a different style than the one required	- Information used is from trusted, rigorous sources - Sources used are deposited in the repository - Sources used are relevant for the statements they support - Sources are cited following the provided style guide	- Reviews or book chapters are generally used to support statements about general context of the disease or topic discussed - Original papers are generally used to support statements about explicit facts and mechanisms that were demonstrated experimentally - Students demonstrate awareness of the difference between primary and secondary literature and select sources according to the type of claim being supported - References are commented, clearly identifying the sources most critical to the topic and those that are simply helpful for expanding the context provided	10%
Bioinformatic database literacy	- Key databases for the disease amongst the ones showed in class are not used - Interpretation of the database results is incorrect	- The selected databases help to resolve the intended problem - Interpretation of database findings is correct - Databases are used correctly, identifying relevant candidates with clear rationale	- Findings are documented with rationale on why they were chosen instead of others - When multiple database records or candidate results are available, the team selects among them using biologically and technically relevant criteria and documents that choice	5%
Disease understanding	- Basic biological understanding of the disease is not obtained - The essential concepts of gene, protein, structure and function are unclear, confused or incorrect	- The origin and consequences of the disease are understood - The main genetic, protein-level, structural and functional causes are identified	- Relevant regulatory or functional pathways are identified when applicable if they exist, as well as important downstream physiological or systemic processes affected - The clinical consequences of the disease are understood, as well as their human impact	10%

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Bioinformatic analysis	- The techniques shown in class are not applied or applied incorrectly - The interpretation of the output of each type of alignment is mistaken or incorrect	- The techniques learned which are relevant to the project topic are applied correctly - The species chosen are justified and useful for the topic of analysis - The output is shared in a clear, understandable format	- Strong understanding of the techniques used is displayed - The chosen species for comparison are relevant for comparison - Where applicable and possible, reference and disease-associated human sequences or variants are incorporated into the analysis - The team understands what conclusions can and cannot reasonably be drawn from each analysis performed	10%
Structural and evolutionary reasoning	- The interpretation of structural findings between the reference and disease-associated states is unclear or not understood - Awareness of the differences in structure and function between humans and other species is not obtained or understood - There is no clear rationale of why the structure is impacting the function	- The team understands the observed results and can explain them and their consequences from a structural perspective - Homologs are clearly identified, and the conservation (or loss of conservation) of the key functional domain(s) is understood alongside its consequences	A strong understanding exists of the relevant protein findings across the structural levels applicable to the project (eg sequence, tertiary and quaternary structure)	10%
Knowledge integration	- The team fails to integrate their evidence and information at the disease level with the experimental findings at gene, protein, structure and function level - The team is unable to explain the difference between impact at gene, protein, structure and function level - Each part of the project is generated in a disconnected way, without identifying and integrating all information towards a clear biological understanding of the problem	- The team is able to understand the disease and how it manifests at genetic, protein, structure and functional levels - The awareness on evolutionary differences between species is explained by understanding their similarities at gene, protein and function levels in an integrated way	- The team can reason about the biological problem in an integrated way and connect it to the clinical reality of patients - The importance of the findings at molecular level is reasoned through a biological understanding of the problem at different scales	5%
Written scientific communication	- Report format is not respected - Argumentation is weak or not explained clearly - Concepts explained are not coherent to the topic of the project or excessive unnecessary information is included - No attribution is given to the work of each team member	- Argumentation is clear, concise and relevant to the topic of the project - Good argumental coherence exists across all sections of the report - The required report format is respected - Attribution of each team member to tasks, report sections and decisions is clear	- Results and discussion are different and well constructed - Abstract respects length and is concise, informative and representative of the project - All concepts mentioned are correctly cited and explained without unreasonably going beyond the project's topic	10%

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Oral scientific communication	- Individual team members show lacking understanding of the content they defend in the written report - Team members fail to explain their findings or statements meaningfully - There is insufficient team member coordination during presentation - Team members show "niche awareness" and only know about the tasks they executed themselves	- All team members show a good understanding of all parts of the project - All claims made in the written report are successfully defended and owned by all team members - Presentation strategy is clear and coordinated - The support used (either existing or not) contributes meaningfully to explaining and defending the project	- Team members show excellent preparation, coordination and division of roles according to their strategy - Questions are addressed correctly within the expected level of understanding, concisely and answers distinguish evidence-supported reasoning from unsupported speculation - All team members display excellent understanding of the project as a whole	10%
Project Execution	- Incorrect identification of the problem to be solved - Incorrect splitting of the problems into tasks to be completed - Most tasks are assigned to several or all team members - Project deliverables are submitted without respecting deadlines	- Research problem is correctly identified - Each part of the research problem is split into actionable tasks - Most tasks are assigned fairly across team members - All deliverables are on time	- Obstacles are identified early, documented and addressed through appropriate adaptation or previously planned mitigation when available - Most tasks are successfully completed within the team's own planned timelines - Decision-making or complex tasks are performed by several members and operational tasks are done by individuals	10%
Project Documentation	- The project formatting requirements are not respected - Commits are done by a single individual - Commits are done in bulk or at deadlines instead of as continuous progress increments - Contribution of different team members varies greatly	- All team members contribute reasonably to the project progress - Commits are incremental during the project development - The repository is structured and organized cleanly during the entire project development	- Commits show clear governance of tasks: operational tasks are performed by individuals and complex tasks receive commits from multiple team members - Commits consistently contain clear, brief comments highlighting the content and rationale of the changes made - The Git history faithfully reflects progressive project development, without substantial work being completed outside version control and subsequently bulk-uploaded	10%

NOTE Descriptors represent anchor points on a continuous 0–1 scale; intermediate scores may be awarded according to the degree to which performance falls between descriptors